The geometric superpotential–FLTZ comparison

Geometric statement

1 Input data and the two endpoints

Fix a based lattice $N \simeq \mathbb{Z}^n$, $n \geq 1$, its dual M, and

$$T = M_{\mathbb{R}}/2\pi M, \qquad Y = M_{\mathbb{R}} \times T, \qquad z = \exp(u + i\theta).$$

Let $Q \subset N_{\mathbb{R}}$ be a full-dimensional lattice polytope containing zero in its interior. Set $V = \mathrm{Vert}(Q)$, assume that the vertices of every facet of Q form a lattice basis of N, and let Σ consist of zero and the cones on all nonempty proper faces of Q. Thus Σ is the smooth complete Fano fan determined by these data and V is exactly its set of primitive ray generators. These are the combinatorial inputs of [2]. Fix $c = (c_v)_{v \in V} \in (\mathbb{C}^*)^V$ and no other monomials:

$$W = W_c = \sum_{v \in V} c_v e^{\langle v,u\rangle + i\langle v,\theta\rangle}, \quad \phi_0 = \tfrac{1}{2}|u|^2, \quad \lambda_0 = \sum_j u_j \, d\theta_j.$$

Use $\iota_Z d\lambda = \lambda$ and

$$\lambda_{\mathbb{C}} = \tfrac{1}{2}(x\,dy - y\,dx), \qquad \lambda_K = \lambda_0 + KW^*\lambda_{\mathbb{C}} \quad (K \geq 0).$$

The angle variables are quotient coordinates: the exponentials and the one-forms $d\theta_j$ descend from $M_{\mathbb{R}}$ to T. They do not require globally defined real-valued angles. With the standard complex structure $J\partial_{u_j} = \partial_{\theta_j}$, W is holomorphic and $d\lambda_K = \sum_j du_j \wedge d\theta_j + KW^*(dx \wedge dy)$ is symplectic: the first summand is positive on (v, Jv) for $v \neq 0$, and the second is nonnegative. Thus Z_K, defined by $\iota_{Z_K} d\lambda_K = \lambda_K$, is well defined for every $K \geq 0$; completeness is a separate conclusion. Completeness means completeness of this Liouville field in both time directions. Finite Liouville type means that inclusion of a compact Liouville domain extends to a primitive-preserving diffeomorphism from its completion onto the given manifold.

Identify $T^*T = T \times N_{\mathbb{R}}$ with primitive $\lambda_{\mathrm{can}} = \sum_j p_j \, d\theta_j$. Choose the Euclidean norm determined by the fixed basis and put

$$D^*T = \{|p| \leq 1\}, \quad S^*T = \{|p| = 1\}, \quad \alpha_{\mathrm{can}} = \lambda_{\mathrm{can}}|_{S^*T}.$$

The target link is the following literal closed subset of S^*T:

$$\begin{aligned} \Lambda_\Sigma^\infty &= \bigcup_{\sigma \in \Sigma,\ \sigma \neq 0} T_\sigma \times (\sigma \cap S^{n-1}), \\ T_\sigma &= \{\theta : \langle v, \theta\rangle \in 2\pi\mathbb{Z} \text{ for every } v \in \sigma \cap N\}. \end{aligned} \tag{1}$$

This specifies the positive-cone, period-2π convention. The FLTZ reference [1, §3.1, equation (8)] provides the formula's name and opposite-cone convention, not a geometric or categorical input. The cotangent lift $(\theta, p) \mapsto (-\theta, -p)$ changes the cone sign. The map $(x, p) \mapsto (2\pi x, p/(2\pi))$ changes period one to period 2π while preserving the cotangent primitive. These maps compare the conic subsets before taking their links in the chosen boundary sections.

2 The geometric data carried by a comparison

Differential-geometric conventions. All manifolds are finite-dimensional, Hausdorff and second countable. Smooth means C^∞, with smooth extension across boundaries and parameter-interval endpoints. A symplectic form is a closed, pointwise nondegenerate two-form. A Liouville domain (D, λ) is a compact smooth manifold with boundary, with $d\lambda$ symplectic and the unique field $\iota_Z d\lambda = \lambda$ strictly outward. Outward means $d\rho(Z) > 0$ for a local defining function $D = \{\rho \le 0\}$ at each boundary point. In dimension zero these conditions hold for the zero form and zero field on a finite set, whose boundary is empty. A contact form on a $(2n-1)$-manifold means $\alpha \wedge (d\alpha)^{n-1}$ is nowhere zero, including $n = 1$.

Definition 2.1 (Witnessed compact pair)**.** *A pair consists of a compact smooth $2n$-manifold with boundary D, a one-form λ with $d\lambda$ symplectic and Z_λ strictly outward at ∂D, a compact smooth $(2n-2)$-manifold with boundary F, a smooth embedding $j : F \hookrightarrow \partial D$, and a smooth positive function $b : \partial D \to \mathbb{R}_{>0}$, such that*

$$\alpha = \lambda|_{\partial D}, \qquad \mu = j^*(b\alpha)$$

makes F a Liouville domain. All smooth data extend across the indicated boundaries. The completion is $\widehat{D} = D \cup_{\partial D} (\partial D \times [0, \infty))$ with end primitive $e^r\alpha$. Its boundary at infinity is represented by ∂D. When another completed ambient space is named, a marking is a specified primitive-preserving diffeomorphism from $\widehat{D}$ onto that space.

The full intrinsic core and its embedded image are

$$\mathrm{Core}(F, \mu) = \bigcap_{t \ge 0} \mathrm{Fl}^{-t}_{Z_\mu}(F), \qquad j(\mathrm{Core}(F, \mu)) \subset \partial D.$$

The flow is computed in the completion of F. Equivalently the core consists of those $x \in F$ for which $\mathrm{Fl}^t_{Z_\mu}(x) \in F$ for every real $t \ge 0$. Here $\mathrm{Fl}^{-t}(F)$ denotes the image under time $-t$. The Liouville collar supplies the smooth gluing in the completion; the field there is ∂_r. Compactness on the interior and this end model give a flow defined for all real times on the completion. When $n = 1$, a ribbon is a compact zero-dimensional manifold; its Liouville field is zero and its core is the entire ribbon.

Definition 2.2 (Permitted comparison steps)**.** *A* pair deformation *is a smooth family of all the data in Definition 2.1 over $[0, 1]$. It may be given on compact smooth domain and ribbon bundles; a chosen trivialization presents it on fixed D and F. Thus λ_t, j_t, b_t and $\mu_t = j_t^*(b_t\lambda_t|_{\partial D})$ are jointly smooth, and both Liouville fields are outward for every t.*

A completion identification *consists of a specified primitive-preserving diffeomorphism $\Psi : \widehat{D} \to \widehat{D}'$ and its induced contact map $\psi : \partial D \to \partial D'$ at infinity, together with a diffeomorphism $h : F \to F'$ satisfying*

$$\Psi^*\lambda' = \lambda, \qquad \psi \circ j = j' \circ h, \qquad h^*\mu' = c_{\mathrm{scale}}\mu \quad (c_{\mathrm{scale}} > 0 \text{ constant}).$$

The boundary map uses Liouville trajectories to compare the chosen boundary sections; it is part of the data, not an arbitrary identification of abstract cores. A primitive-preserving map intertwines the Liouville fields and hence induces this map at infinity. More explicitly, for $x \in \partial D$, follow the escaping trajectory $\Psi(\mathrm{Fl}^r_Z(x))$ to the unique representative on $\partial D'$; this representative is $\psi(x)$. Equivalently, for large r its end coordinates are $(\psi(x), r + s(x))$ for a smooth real function s. Primitive preservation gives $\psi^*\alpha' = e^{-s}\alpha$.

Write $\mathcal{P} \rightsquigarrow \mathcal{Q}$ for a finite sequence of these steps and their reverses, with an explicit finite list of endpoint tuples. Consecutive steps share their endpoint tuple literally; any change of presentation must be recorded as a permitted step with its maps. If the witnessing forms do not match, an actual pair deformation must be inserted. Neither replacement by a singular subset nor exchange of two ribbons merely because their embedded cores agree is a permitted step. These are deliberate geometric requirements, stronger than the minimum needed for categorical sufficiency.

Remark 2.3. The positive factor b is essential data: arbitrary multiplication of a ribbon primitive by a positive function need not preserve its symplectic differential. No assertion here replaces λ on the ambient manifold by $b\lambda$. A bare exact symplectomorphism with uncontrolled behavior at infinity would also be insufficient for Definition 2.2. Primitive preservation is the stronger condition available for the completion identifications intended here.

Definition 2.4 (Ambient Maslov datum and transport)**.** *For a symplectic manifold (X, ω) choose an ω-compatible almost complex structure J and write $K_J = \Lambda^n_{\mathbb{C}}(T^{1,0}X)^*$ for its complex canonical line bundle. A Maslov datum is a homotopy class of nowhere-vanishing sections of $K_J^{\otimes 2}$. All sections and homotopies in this definition are smooth; section homotopies remain nowhere vanishing and homotopies of complex structures remain compatible with the relevant symplectic form. Compatibility means $\omega(Jv, Jw) = \omega(v, w)$ and $\omega(v, Jv) > 0$ for $v \neq 0$. Changing J through compatible structures identifies these classes; the space of compatible structures is contractible.*

On a pair deformation the datum is carried on the completed ambient family: choose a smooth compatible J_t and a nowhere-vanishing section of the resulting squared canonical line bundle over the whole family. Its restrictions specify the endpoint classes. For a completion identification Ψ, a graded lift is a homotopy from the source section to the pullback of the target section, after identifying the compatible complex structures by a homotopy. A graded chain consists of a chain and these extension and homotopy data, agreeing at junctions. This definition asserts neither existence of a datum on every symplectic manifold nor the prescribed endpoint class for every possible chain.

On Y the initial datum is $[\Omega^{\otimes 2}]$, where $\Omega = \bigwedge_j (du_j + i\, d\theta_j)$ and the standard J above is compatible with every $d\lambda_K$. On T^*T use $J_{\mathrm{can}}\partial_{p_j} = \partial_{\theta_j}$ and the class of $\left(\bigwedge_j (dp_j + i\, d\theta_j)\right)^{\otimes 2}$. Its squared phase on a vertical Lagrangian plane is one; we call this the vertical-phase-zero datum. No grading is assigned to the singular subset Λ_Σ^∞ or to individual branes here.

3 The specified initial ribbon and its admissible choices

The following end bounds specify admissibility without assuming fullness or the desired comparison. Put

$$m_v = c_v e^{\langle v,u\rangle + i\langle v,\theta\rangle}, \qquad G = \sum_v m_v v,$$

$$a = \overline{G}^T G, \qquad b_0 = u \cdot G, \qquad Q_W = |u|^2 a - |b_0|^2 + \tfrac{1}{2}\,\mathrm{Re}(\overline{W} b_0).$$

An ambient end bound $A_{\mathrm{amb}} > 0$ satisfies $a > 0$ and $Q_W > 0$ on $\{\phi_0 \geq A_{\mathrm{amb}}\}$. Write $M(A_{\mathrm{amb}}) = \max_{\{\phi_0 \leq A_{\mathrm{amb}}\}} |W|$; this maximum exists on a nonempty compact set. For $R > \max(1, M(A_{\mathrm{amb}}))$, define the ratio set

$$S_R = \left\{ \frac{|b_0(y)|}{a(y)|W(y)|} : y \in Y,\ |W(y)| \geq R \right\}.$$

Its denominators are positive, since $|W| \geq R$ implies $\phi_0 > A_{\mathrm{amb}}$. When S_R is nonempty and bounded above, put

$$K_0(R) = 1 + 2 \sup S_R. \tag{2}$$

Existence of an ambient end bound, nonemptiness of S_R, and boundedness of S_R are conclusions required of the target. In particular $K_0(R)$ is this specified real number, at least one, rather than an independently chosen threshold. A formalization using a total real supremum must retain the nonemptiness and boundedness assertions explicitly. A fiber end bound $A_{\mathrm{fib}}(R) \geq A_{\mathrm{amb}}$ satisfies

$$a > 0, \qquad |u|^2 - \frac{|b_0|^2}{a} > 0 \quad \text{on } \{\phi_0 \geq A_{\mathrm{fib}}(R),\ |W| \leq R\}. \tag{3}$$

Its existence is likewise part of the target. When $n = 1$, the expression $|u|^2 - |b_0|^2/a$ is zero wherever $a > 0$. Consequently a valid fiber end bound makes the set on which (3) is required empty. Such a bound is possible because $W = c_+ e^{u+i\theta} + c_- e^{-u-i\theta}$ is proper in this case: the triangle inequality shows $|W| \to \infty$ as $u \to \pm\infty$. Do not replace (3) by a positivity claim on a nonempty end in dimension one.

For such bounds and a fixed $K \geq K_0(R)$ choose

$$A_- > A_{\mathrm{fib}}(R), \qquad A > A_-.$$

Use a smooth convex function $m_\varepsilon : \mathbb{R}^2 \to \mathbb{R}$ with nonnegative partial derivatives whose sum is one, satisfying

$$\max(x, y) \leq m_\varepsilon(x, y) \leq \max(x, y) + \varepsilon, \qquad m_\varepsilon(x, y) = \max(x, y) \text{ if } |x - y| \geq \varepsilon.$$

Define

$$\begin{aligned} E_\varepsilon &= \{m_\varepsilon(\phi_0 - A,\ |W|^2 - R^2) \leq 0\}, \\ F_R^- &= \{W = R,\ \phi_0 \leq A_-\}, \qquad \mu_R = \lambda_0|_{F_R^-}. \end{aligned} \tag{4}$$

Definition 3.1 (Admissible rounding)**.** *For these fixed R, K, A_-, A, a rounding consists of a real ε and a profile m_ε satisfying all the smoothness, convexity, derivative and equality conditions above. It is admissible when $0 < \varepsilon < A - A_-$ and, at every point of the explicitly defined band*

$$B_\varepsilon = \{-\varepsilon < \phi_0 - A \leq 0, \quad -\varepsilon < |W|^2 - R^2 \leq 0\},$$

one has

$$d(|W|^2)(Z_K) > 0, \qquad d\phi_0 \wedge d(|W|^2) \neq 0.$$

The second condition means linear independence of the two real covectors. It is allowed to hold vacuously when B_ε is empty. These conditions do not assume a completion marking, core equality, or comparison chain. Existence of admissible roundings and the asserted regularity of their domains are conclusions of the target.

For orientation, set $E_0 = \{\phi_0 \leq A,\ |W| \leq R\}$. The profile conditions imply $E_\varepsilon \subset E_0$ and $E_0 \setminus E_\varepsilon \subset B_\varepsilon$. Indeed $m_\varepsilon(x + t, y + t) = m_\varepsilon(x, y) + t$, by the sum of the partial derivatives. If $x \leq -\varepsilon$ and $y \leq 0$, monotonicity gives $m_\varepsilon(x, y) \leq m_\varepsilon(-\varepsilon, 0) = 0$; interchange x, y for the other case. The portion of the zero level on which $m_\varepsilon > \max$ also lies in B_ε. To see the strict band bounds there, use $m_\varepsilon(x, y) = x + g(y - x)$, where g is nondecreasing, $g(\varepsilon) = \varepsilon$, and $g'(\varepsilon) = 1$. Hence $g(s) < \varepsilon$ for $s < \varepsilon$, whereas $g(s) = s$ for $s \geq \varepsilon$. If $x \leq -\varepsilon$ and $y < 0$, these two cases both give $m_\varepsilon(x, y) < 0$; the other case is symmetric. Unrounded faces need not lie in B_ε. The band is disjoint from F_R^- because $\phi_0 > A - \varepsilon > A_-$ there. For $n = 1$ it is empty, $E_\varepsilon = \{|W| \leq R\}$, and $F_R^- = W^{-1}(R)$. On the unrounded face containing the ribbon, its inclusion j_R has

$$j_R^*(\lambda_K|_{\partial E_\varepsilon}) = \mu_R,$$

since W is constant on F_R^-. The initial witnessing factor is one.

4 The geometric endpoint theorem

Target theorem 4.1 (Geometric endpoint with specified bounds and grading)**.** *For every based lattice, polytope and coefficient tuple of Section 1, the following assertions hold with the nested quantifiers stated below.*

(G1) *For every real $K \geq 0$, (Y, λ_K) is complete and of finite Liouville type. There is an ambient end bound A_{amb} as in Section 3.*

(G2) *The ambient bound in (G1) can be chosen together with $R_0 > \max(1, M(A_{\mathrm{amb}}))$ so that, for every real $R \geq R_0$, S_R is nonempty and bounded above, and at least one fiber end bound $A_{\mathrm{fib}} \geq A_{\mathrm{amb}}$ exists. For every such R, for every valid fiber end bound (relative to this A_{amb}), for every real $K \geq K_0(R)$, and for every pair of levels $A_- > A_{\mathrm{fib}}$ and $A > A_-$, there exists an admissible rounding as specified in Section 3. For every such choice of levels and rounding, (4) defines a witnessed pair with nonempty ribbon. In particular R is a regular value of W and the full fiber is smooth. Inclusion extends to specified primitive-preserving diffeomorphisms*

$$\widehat{(E_\varepsilon, \lambda_K)} \longrightarrow (Y, \lambda_K), \qquad \widehat{(F_R^-, \mu_R)} \longrightarrow (W^{-1}(R), \lambda_0|_{W^{-1}(R)}).$$

These markings are induced by the corresponding Liouville flows and the given inclusions. Thus the ribbon is a truncation of the whole fiber. The fiber marking preserves the source theorem's fullness assertion; the categorical deduction itself only uses the ambient marking.

(G3) *For each of those initial pairs there exist a compact smooth ribbon F_Σ, an embedding $j_\Sigma : F_\Sigma \hookrightarrow S^*T$, and a smooth positive $b_\Sigma : S^*T \to \mathbb{R}_{>0}$ such that*

$$\mu_\Sigma = j_\Sigma^*(b_\Sigma \alpha_{\mathrm{can}})$$

is Liouville, and there is a finite chain

$$(E_\varepsilon, \lambda_K, F_R^-, j_R, 1) \rightsquigarrow (D^*T, \lambda_{\mathrm{can}}, F_\Sigma, j_\Sigma, b_\Sigma)$$

in the sense of Definition 2.2. The endpoint data and this chain are chosen together with the grading data specified below. The source marking is the one in (G2); the terminal marking is the standard completion of D^*T with T^*T. The endpoint satisfies the equality of embedded subsets

$$j_\Sigma\big(\mathrm{Core}(F_\Sigma, \mu_\Sigma)\big) = \Lambda_\Sigma^\infty \subset S^*T. \tag{5}$$

The same chain *is equipped with ambient Maslov data and graded lifts in the sense of Definition 2.4. At its source the datum is the pullback of $[\Omega^{\otimes 2}]$ by the marking in (G2). At its endpoint, under the standard cotangent marking, the transported datum is homotopic to the vertical-phase-zero datum of Definition 2.4. This is an existential requirement on the chosen chain, not a claim about every chain between these endpoints.*

Quantifier order. For each input, all-K completeness and finite type are asserted first. There exist A_{amb} and R_0 satisfying the stated conditions. For every $R \geq R_0$, assert nonemptiness and boundedness of S_R and existence of a fiber end bound; then define $K_0(R)$ by (2). For *every* valid fiber end bound, every $K \geq K_0(R)$, and every $A_- > A_{\mathrm{fib}}$, $A > A_-$, assert existence of an admissible rounding. For *every* admissible rounding, assert the witnessed pair and both specified markings, followed by one existential package containing the endpoint ribbon, witness, chain, embedded-core equality, and grading extensions and endpoint homotopy. The existence clauses precede the

universal admissibility clauses and are not replaced by them. The endpoint ribbon and the chain need not be independent of R, K or the choices; their embedded core is always the fixed set (1). Item (G1) is asserted also below the ribbon threshold. The target assumes neither a strongly compatible potential, a full-core identification, nor the desired chain as input. Their construction belongs to its proof.

References

[1] B. Fang, C.-C. M. Liu, D. Treumann, and E. Zaslow, *T-duality and homological mirror symmetry of toric varieties*, arXiv:0811.1228v4. https://arxiv.org/html/0811.1228v4.

[2] *Fukaya categories of toric superpotentials and the FLTZ stop: An explicit far-fiber comparison for smooth Fano fans*, manuscript.